

From the Quantum Phase estimation algorithm to the HHL and QAOA algorithms

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Outline

1. Quantum Phase Estimation (QPE)
2. Connection to Hamiltonian simulation
3. Connection to Variational Quantum Eigensolvers (VQE)
4. Introducing the HHL algorithm

See also separate jupyter-notebooks at <https://github.com/CompPhysics/QuantumComputingMachineLearning/tree/gh-pages/doc/pub/week11/ipynb/week11.ipynb>

Quantum Phase Estimation: Problem Statement, reminder from Lecture March 25

Suppose U is a unitary operator and $|\psi\rangle$ is one of its eigenstates:

$$U|\psi\rangle = e^{2\pi i\phi} |\psi\rangle, \quad 0 \leq \phi < 1.$$

The goal of Quantum Phase Estimation (QPE) is to estimate the phase ϕ .

Equivalently, QPE extracts the eigenvalue of U in phase form.

Why QPE Matters

QPE is central because many physical problems can be mapped to eigenvalue estimation.

Examples:

- ▶ energy estimation in quantum chemistry,
- ▶ eigenvalue problems in many-body physics,
- ▶ Hamiltonian simulation,
- ▶ Shor's algorithm and period finding.

If we can encode a Hamiltonian into a unitary, then QPE can extract its eigenvalues.

Registers Used in QPE

QPE uses two registers:

- ▶ a control register of n qubits, initialized in $|0\rangle^{\otimes n}$,
- ▶ a system register, initialized in the eigenstate $|\psi\rangle$.

So the initial state is

$$|0\rangle^{\otimes n} |\psi\rangle .$$

The control register stores the estimated phase bits.

Step 1: Create a Uniform Superposition

Apply Hadamard gates to all control qubits:

$$|0\rangle^{\otimes n} \mapsto \frac{1}{2^{n/2}} \sum_{k=0}^{2^n-1} |k\rangle.$$

So the total state becomes

$$\frac{1}{2^{n/2}} \sum_{k=0}^{2^n-1} |k\rangle |\psi\rangle.$$

This prepares the control register as a coherent superposition of all binary numbers k .

Step 2: Controlled Powers of U

Now apply controlled powers of U :

$$U^{2^0}, U^{2^1}, U^{2^2}, \dots, U^{2^{n-1}}.$$

Each control qubit determines whether the corresponding power is applied.

Because $|\psi\rangle$ is an eigenstate,

$$U^{2^j} |\psi\rangle = e^{2\pi i 2^j \phi} |\psi\rangle.$$

Hence each controlled unitary writes phase information into the control register.

Step 2: State After Controlled Powers

If the control register basis state is $|k\rangle$, then the controlled operations apply U^k , giving

$$U^k |\psi\rangle = e^{2\pi i k \phi} |\psi\rangle .$$

Therefore the combined state becomes

$$\frac{1}{2^{n/2}} \sum_{k=0}^{2^n-1} e^{2\pi i k \phi} |k\rangle |\psi\rangle .$$

The system register factors out, and the control register now contains the phase information.

Key Observation

The control register is in the state

$$\frac{1}{2^{n/2}} \sum_{k=0}^{2^n-1} e^{2\pi i k \phi} |k\rangle ,$$

which is exactly a Fourier-type state.

This is why the inverse QFT is the correct next step: it converts this phase-encoded superposition into a computational basis approximation of ϕ .

Step 3: Apply the Inverse QFT

Applying QFT^{-1} to the control register gives

$$\text{QFT}^{-1} \left(\frac{1}{2^{n/2}} \sum_{k=0}^{2^n-1} e^{2\pi i k \phi} |k\rangle \right).$$

If

$$\phi = \frac{m}{2^n}$$

exactly for some integer m , then this becomes

$$|m\rangle.$$

So the control register stores the exact n -bit binary expansion of ϕ .

Exact QPE Case

Suppose

$$\phi = 0.\phi_1\phi_2\dots\phi_n = \frac{m}{2^n}.$$

Then after inverse QFT, the control register becomes

$$|\phi_1\phi_2\dots\phi_n\rangle.$$

Thus measuring the control register returns the exact binary digits of ϕ .

This is the cleanest case of QPE.

Step-by-Step Derivation with Binary Fractions

Let

$$\phi = 0.\phi_1\phi_2 \dots \phi_n \dots$$

Then one can show that the control-register state after controlled powers factorizes as

$$\frac{1}{2^{n/2}} \left(|0\rangle + e^{2\pi i 2^{n-1} \phi} |1\rangle \right) \otimes \left(|0\rangle + e^{2\pi i 2^{n-2} \phi} |1\rangle \right) \otimes \dots \\ \dots \otimes \left(|0\rangle + e^{2\pi i \phi} |1\rangle \right).$$

Because

$$2^{n-1} \phi = \phi_1.\phi_2\phi_3 \dots,$$

only the fractional parts matter in the phases, so each qubit encodes one successive binary suffix.

How the Bits Emerge

Using the binary expansion of ϕ , the control-register state becomes

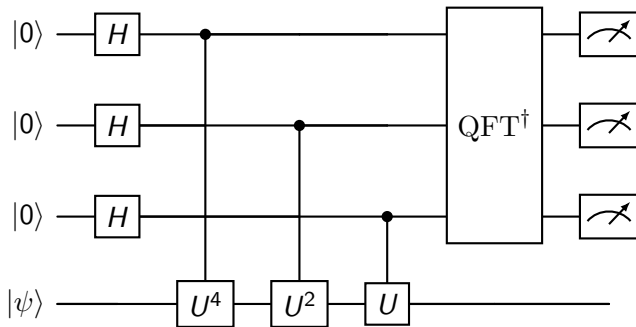
$$\frac{1}{2^{n/2}} \left(|0\rangle + e^{2\pi i 0.\phi_n} |1\rangle \right) \otimes \left(|0\rangle + e^{2\pi i 0.\phi_{n-1}\phi_n} |1\rangle \right) \otimes \dots \\ \dots \otimes \left(|0\rangle + e^{2\pi i 0.\phi_1\phi_2\dots\phi_n} |1\rangle \right),$$

which has exactly the structure of a QFT state.

Therefore applying QFT^{-1} yields the bit string

$$\phi_1\phi_2\dots\phi_n.$$

QPE Circuit



This example shows a 3-qubit control register and one system register.

Worked Example: $\phi = 5/8$

Suppose

$$\phi = \frac{5}{8} = 0.101.$$

Then the ideal QPE output on a 3-qubit control register is

$$|101\rangle.$$

Indeed,

$$e^{2\pi i \phi} = e^{2\pi i (5/8)}.$$

After the controlled powers and inverse QFT, the control register collapses to the bit string 101 with probability 1.

When ϕ Is Not Exactly Representable

If ϕ is not exactly of the form $m/2^n$, then QPE does not return one exact bit string.

Instead, the output is a probability distribution peaked near the best n -bit approximation:

$$\phi \approx \frac{m}{2^n}.$$

The measurement outcome gives the nearest binary approximation with high probability.

Precision of QPE

With n control qubits, QPE resolves phases to approximately

$$\Delta\phi \sim \frac{1}{2^n}.$$

Thus the precision improves exponentially with the number of control qubits.

This exponential precision is one of the main reasons QPE is so powerful.

From Phases to Energies

Suppose H is a Hamiltonian with eigenstate $|E\rangle$:

$$H|E\rangle = E|E\rangle.$$

Define the time-evolution operator

$$U(t) = e^{-iHt}.$$

Then

$$U(t)|E\rangle = e^{-iEt}|E\rangle.$$

Comparing with

$$U|\psi\rangle = e^{2\pi i\phi}|\psi\rangle,$$

we identify

$$\phi = -\frac{Et}{2\pi}.$$

Thus QPE can estimate energies if we can implement e^{-iHt} .

Connection to Hamiltonian Simulation

QPE requires controlled applications of

$$U^{2^j} = e^{-iHt2^j}.$$

So QPE depends directly on the ability to perform Hamiltonian simulation.

In practice this may be done using:

- ▶ Trotter-Suzuki decompositions,
- ▶ linear-combination-of-unitaries methods,
- ▶ qubitization,
- ▶ problem-specific simulation techniques.

Thus:

Hamiltonian simulation + QPE $\Rightarrow$ energy estimation.

Physical Interpretation

QPE measures the dynamical phase accumulated by an energy eigenstate under time evolution.

The algorithm turns the phase

$$e^{-iEt}$$

into a bit string encoding the energy.

So one can view QPE as a highly precise interferometric energy-measurement protocol.

Why QPE Is Important in Quantum Chemistry

In quantum chemistry and many-body physics, the main goal is often to compute eigenenergies of a Hamiltonian.

If we can prepare an approximate eigenstate with sufficient overlap with the true ground state, then QPE can project onto the exact eigenvalue with high precision.

This makes QPE a natural route to:

- ▶ ground-state energies,
- ▶ excited-state energies,
- ▶ spectral properties.

Connection to VQE

The Variational Quantum Eigensolver (VQE) and QPE aim at similar physics goals, but in different ways.

VQE:

- ▶ variational,
- ▶ hybrid quantum-classical,
- ▶ optimized by minimizing

$$E(\theta) = \langle \psi(\theta) | H | \psi(\theta) \rangle ,$$

- ▶ suitable for noisy near-term devices.

QPE:

- ▶ not variational,
- ▶ requires coherent controlled time evolution,
- ▶ provides direct eigenvalue estimation,
- ▶ suited to fault-tolerant quantum computers.

Note on variational theorem

QPE is a non-variational, projector-based approach. It acts directly on the Hilbert space to extract eigenvalues. Because QPE directly computes the eigenvalues (within precision limits) of a Hamiltonian via phase estimation (often using Trotter decomposition), it is not restricted to providing an "upper bound" based on a trial wavefunction.

How VQE and QPE Complement Each Other

A common long-term strategy is:

1. Use a variational method such as VQE to prepare a good approximate eigenstate

$$|\psi_{\text{trial}}\rangle$$

with large overlap with the true eigenstate.

2. Use that state as the input to QPE.
3. QPE then refines the energy estimate to high precision.

So VQE can be viewed as a state-preparation front end for QPE.

Overlap Requirement

If the initial state is

$$|\psi_{\text{trial}}\rangle = \sum_j c_j |E_j\rangle,$$

then QPE outputs E_j with probability

$$|c_j|^2.$$

Therefore the success probability depends on the overlap with the desired eigenstate.

This is exactly why improved ansätze in VQE are also valuable for QPE-based workflows.

QPE vs VQE in Practice

VQE

- ▶ shallow circuits,
- ▶ many repeated measurements,
- ▶ classical optimization loop,
- ▶ noise tolerant,
- ▶ limited by optimization difficulty and shot cost.

QPE

- ▶ deep coherent circuits,
- ▶ requires controlled e^{-iHt} ,
- ▶ very precise,
- ▶ ideal for fault-tolerant hardware,
- ▶ often viewed as the long-term gold standard for energy estimation.

Semiclassical Interpretation

One may interpret QPE as repeatedly extracting successive binary digits of a phase.

The inverse QFT acts globally, but conceptually it decodes the phase one bit at a time.

This bitwise structure is one reason QPE is so elegant mathematically.

Implementation

A implementation of the QPE typically proceeds as follows:

1. choose the number of control qubits,
2. initialize the system register in an eigenstate or approximate eigenstate,
3. apply Hadamards to the control register,
4. apply controlled U^{2^j} ,
5. apply inverse QFT,
6. measure the control register,
7. interpret the measurement as an estimate of ϕ .

If the notebook also implements the QFT explicitly, it will usually reuse the same Hadamard + controlled-phase + swap pattern.

Summary of the Full Picture

$$\text{Hamiltonian } H \longrightarrow U(t) = e^{-iHt} \longrightarrow \text{QPE} \longrightarrow E$$

The QFT is the phase-decoding engine inside QPE.

VQE and QPE are complementary:

- ▶ VQE is a near-term variational energy estimator,
- ▶ QPE is a long-term high-precision eigenvalue estimator,
- ▶ VQE can provide input states for QPE.

Summary

We have seen that:

- ▶ the QFT maps

$$|x\rangle \mapsto \frac{1}{\sqrt{N}} \sum_k e^{2\pi i x k / N} |k\rangle,$$

- ▶ QPE uses controlled powers of a unitary and the inverse QFT to extract eigenphases,
- ▶ for Hamiltonian evolution

$$U(t) = e^{-iHt},$$

QPE becomes an energy-estimation algorithm,

- ▶ VQE and QPE address related eigenvalue problems with different hardware requirements.

Suggestions for the second project

- ▶ Continuation of project 1 with more realistic systems and using adaptive VQE on actual (real) quantum computers
- ▶ Implementation and studies of the Quantum Approximate Optimization Algorithm (QAOA): Simulation of linear algebra systems on quantum computers, solving for example differential equations
- ▶ Implementation of the HHL algorithm (needs QPE)
- ▶ Applications and implementations of quantum machine learning algorithms
- ▶ Implement quantum neural networks for PINNs
- ▶ Detailed quantum chemistry applications.

Suggestions for the second project

- ▶ Studies of entanglement and physical realization of quantum gates and circuits
- ▶ Implementing Shor's algorithm or other algorithms
- ▶ Parallelizing the Variational Quantum Eigensolver with multi-GPU scaling, see <https://arxiv.org/abs/2601.09951>
- ▶ Implementing and comparing the quantum phase estimation algorithm with VQE, see for example recent QPE paper at "<https://arxiv.org/abs/2601.05788>"
- ▶ Studies of hyperparameters in VQE, see for example <https://arxiv.org/abs/2601.16679>
- ▶ Implementing unitary coupled cluster theory on a quantum computer, see <https://arxiv.org/abs/2501.15893>

What is the HHL algorithm?

The HHL algorithm—named after Aram Harrow, Avinatan Hassidim, and Seth Lloyd—is one of the central quantum algorithms for solving linear systems of equations.

The core idea is to solve a linear system

$$A\mathbf{x} = \mathbf{b}.$$

What is the QAOA algorithm?, to be discussed April 15

QAOA stands for the Quantum Approximate Optimization Algorithm

Each part of the name reflects something essential about the method:

- ▶ Quantum: it runs on a quantum computer, using qubits and quantum gates.
- ▶ Approximate: it typically finds near-optimal solutions rather than exact ones.
- ▶ Optimization: it is designed to solve optimization problems (like MaxCut, scheduling, portfolio selection).
- ▶ Algorithm: it is a well-defined hybrid quantum–classical procedure.

QAOA is a variational quantum algorithm that uses a parameterized quantum circuit to search for good solutions to hard optimization problems.

HHL: Linear Systems Everywhere

- ▶ Solve:

$$A\mathbf{x} = \mathbf{b}$$

- ▶ Appears in:
 - ▶ PDEs
 - ▶ Machine learning and quantum machine learning
 - ▶ Many-body physics
- ▶ Classical cost: $\mathcal{O}(N^3)$ (dense case)

Quantum Goal

- ▶ Encode vector as quantum state:

$$|b\rangle = \sum_i b_i |i\rangle$$

- ▶ Goal:

$$|x\rangle \propto A^{-1}|b\rangle$$

- ▶ Output is a quantum state, not a classical vector

Eigenbasis Representation

$$A|u_j\rangle = \lambda_j|u_j\rangle$$

$$|b\rangle = \sum_j \beta_j |u_j\rangle$$

Formal Solution

$$A^{-1}|b\rangle = \sum_j \frac{\beta_j}{\lambda_j} |u_j\rangle$$

- Problem reduces to eigenvalue inversion